

Wael Hammali

Software Engineering student at National Engineering School of Tunis (ENIT)

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Tunis, Tunisia

PROFILE

Third-year software engineering student at ENIT specializing in Artificial Intelligence, Computer Vision and Cybersecurity, seeking opportunities to enrich my background and advance my career in research.

EDUCATION

- National Engineering School of Tunis (ENIT)** *September 2024 - Present*
B.Eng. in Software Engineering *Tunis, Tunisia*
- Preparatory Institute for Engineering Studies of Monastir (IPEIM)** *September 2021 - July 2023*
Preparatory cycle for engineering studies, specializing in Physics and Technology – High Honors (Rank 78) *Monastir, Tunisia*

ACADEMIC PROJECTS

- From Network Diagrams to Deployable Cloud Infrastructure: A Multimodal AI- Based Approach** *ENIT, Tunisia*
Year 2 Final Project *2025*
 - Engineered a full-stack infrastructure migration tool with a GUI accepting architecture diagrams or text prompts; applied fine-tuned YOLOv8 for network component detection and RAG (Retrieval-Augmented Generation) to map physical infrastructure to cloud equivalents, auto-generating production-ready Terraform & Ansible IaC configurations.
 - Stack:** Python · YOLOv8 · RAG · OpenCV · Terraform · Ansible · LLM
- IntelForge Scanning Framework** *Keystone Groupe, Tunisia*
AI and CyberSecurity project *2026*
 - Built an autonomous penetration testing framework combining passive OSINT (FinalRecon — Whois, DNS, SSL, Shodan, Wayback Machine) with active network reconnaissance (Nmap TCP/UDP, directory & vhost fuzzing) and a 4-stage AI pipeline powered by Groq LLMs and DeepSeek R1 reasoning. The tool automatically progresses from target discovery to structured exploit intelligence reports, drastically reducing manual enumeration time .
 - Stack:** Python · Bash Scripting · LLMs
- Aerial Object Detection with YOLOv8** *ENIT, Tunisia*
Computer Vision project *2025*
 - Designed a dual-model YOLOv8 pipeline to address severe class imbalance in aerial imagery: one model dedicated to tree detection and a second to cars and buildings, integrated with a Gradio web interface.
 - Stack:** Python · YOLOv8 · OpenCV · Fine-tuning · Roboflow
- VoidHawk Penetration Testing Framework (Extended fork of Watchtower)** *Keystone Groupe, Tunisia*
AI and CyberSecurity project *2026*
 - Extended the open-source Watchtower framework with LLM-powered vulnerability validation, a RAG-based memory system (ChromaDB + Sentence-Transformers) for persistent cross-session knowledge retention, and an automated vulnerability comparison and severity-ranking engine. The system orchestrates 6 specialized AI agents (Planner, Worker, Cleaner, Analyst, Logic Analysis, Validator) via a LangGraph state machine and produces CVSS-scored security reports in PDF, HTML, and Markdown formats.
 - Stack:** Python · LangGraph · Groq · OpenAI
- Data Analysis Techniques in Data Lakes** *ENIT, Tunisia*
Year 1 Final Project *2024*
 - Explored data analysis techniques and architectures within Data Lakes in the context of Big Data, focusing on the management of structured, semi-structured, and unstructured data. Studied data preprocessing, modeling, visualization, and governance approaches, while analyzing technologies such as Apache Hadoop, Apache Spark, Databricks, Kafka, and Tableau for scalable data processing and analysis.

RESEARCH AND PROFESSIONAL EXPERIENCE

• Automated Camera Calibration for Augmented Reality in Soccer Analytics

Research Project – SympactAI, 2nd Edition

TAIS NextGen, Tunisia

June 2026 - August 2026

- Participated in the SympactAI – Second Edition research program, contributing to an AI-driven project focused on augmented reality integration in soccer analytics. Led the development of a jersey number recognition module using computer vision and image processing techniques to automatically identify players from live and recorded soccer footage, enhanced with Real-ESRGAN super-resolution to improve accuracy on low-resolution and motion-blurred broadcast footage. Gained practical experience in building real-world AI pipelines while collaborating within a multidisciplinary team on a sports analytics application.

Stack: PyTorch · OpenCV · Real-ESRGAN · Vitpose · PARSeq · Computer Vision · Image Processing

• Master of Research in Communication Systems (SYSCOM)

Research Master – École Nationale d'Ingénieurs de Tunis (ENIT)

Tunis, Tunisia

2026 – Present

- Research-oriented program focused on advanced communication and intelligent systems, covering telecommunications, wireless communications, computer networks, signal processing, RF and antenna systems, IoT, and Artificial Intelligence and Machine Learning for communication technologies. Emphasis on scientific research, system modeling, simulation, and the development of innovative intelligent solutions.

Focus: Artificial Intelligence · Machine Learning · Telecommunications · Computer Networks · Signal Processing · IoT · Wireless Communications · RF Systems

From Network Diagrams to Deployable Cloud Infrastructure: A Multimodal AI-Based Approach

Tunis, Tunisia

Research Manuscript, RISC Laboratory, ICT Department – ENIT, University of Tunis El Manar

2025 – Oct. 2026

- Co-authored with Dhiaeddine Barhoumi, supervised by Mme. Wafa Meftah (full technical details under Projects).

Outcome: Paper accepted at IEEE AIDIST 2026 – "L'intelligence artificielle pour une transition numérique et durable", 5th FCIERI Forum (29–31 Oct. 2026, Tunis).

AI-Powered Penetration Testing Command Framework (DarkIntel)

Keystone Groupe, Tunisia

Engineering Internship

June 2026 - July 2026

- Architected DarkIntel, a three-stage AI pentesting pipeline connecting two independently built frameworks plus a synthesis layer. IntelForge performs passive OSINT and active network reconnaissance, producing structured findings through a multi-stage LLM pipeline. These findings are injected directly into VoidHawk's RAG memory (ChromaDB + Sentence-Transformers), letting its Planner agent skip discovery and start straight at the validation phase. VoidHawk then drives 6 specialized agents through a LangGraph state machine to confirm findings and assign CVSS scores. Finally, Dragon LLM a synthesis stage that reads only the confirmed findings and the actual commands executed, and generates an evidence-grounded exploitation narrative, citing exclusively commands that were verifiably run.

Stack: Python · LangGraph · LangChain · Ollama · Claude (Anthropic API) · ChromaDB · Sentence-Transformers · SQLite

Network & Security Infrastructure Internship

TAV Airports, Enfidha, Tunisia

Engineering Internship

July 2025 - August 2025

- Audited the parking camera and network infrastructure including cabling, switches, firewalls, and servers to identify security vulnerabilities and reliability issues, and contributed to a workaround that maintained system operation during a fiber-optic outage. Produced an infrastructure report recommending hardware upgrades, IoT sensors, and AI-driven computer vision solutions such as license plate recognition and anomaly detection to improve security and operational intelligence. Also proposed strict data access policies to ensure captured data remained protected and restricted from unauthorized access, including internal staff.

CERTIFICATIONS

[1] Wael Hammali (2025). **Certified Associate Penetration Tester (CAPT)**. Hackvoiser, December 2025.

[2] Wael Hammali (2025). **CCNA: Introduction to Networks**. Cisco Networking Academy, January 2026.

[3] Wael Hammali (2026). **CCNA: Switching, Routing, and Wireless Essentials**. Cisco Networking Academy, 2026.

TECHNICAL SKILLS

- | | |
|--|--|
| • Artificial Intelligence / Machine Learning | • Python / Node.js / SQL / MongoDB |
| • Computer Vision / Retrieval-Augmented Generation | • Burp Suite / Metasploit / Wireshark / Bash scripting |
| • LLM fine-tuning / Prompt engineering | • Cybersecurity / Penetration testing / OSINT |
| • Federated Learning / Infrastructure automation | |

SOFT SKILLS

- | | |
|---|-----------------------|
| • Problem solving | • Communication |
| • Critical thinking / analytical thinking | • Time management |
| • Teamwork | • Continuous learning |
| • Adaptability | • Autonomy |

EXTRACURRICULAR ACTIVITIES

- **Active Member** 2024 - Present
Securinets ENIT – Cybersecurity Club
 - Participating in cybersecurity activities, technical workshops, CTF competitions, and security-related events.
- **Golden Member & Event Organizer** 2024 - Present
G2FOSS ENIT – Software Club
 - Organized the **League of Coders**, a major problem-solving and competitive programming event in Tunisia, contributing to event planning, coordination, and participant engagement; later recognized as a Golden Member of the club.
- **Organizing Member** 2024 - Present
FabLab ENIT
 - Helped organize **Aero-Fab**, a robotics competition featuring junior line-follower robots and all-terrain robots, contributing to event coordination and technical activities.
- **Participant – Injaz El Arab Competition** 2025
National Engineering School of Tunis (ENIT)
 - Participated in the Injaz El Arab competition with a sustainable project focused on developing thermal insulation panels from natural materials, strengthening skills in teamwork, project management, problem-solving, and sustainable innovation.
- **Participant – CyberHorizon Hackathon (5G Edition)** 2025
Securinets FST
 - Competed in the CyberHorizon Hackathon with team *EL Kes7inn*, tackling a cybersecurity challenge under time constraints and **ranked among the winning teams**, strengthening skills in teamwork, problem-solving, and applied cybersecurity.

LANGUAGES

- **Arabic**
Native language
- **French**
B2 level
- **English**
B2 level